Predicting Human Reading Times: Causal vs. Masked Language Models on Syntactic Ambiguity

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Abstract

Neural language models have become standard tools for predicting human reading times, yet the relationship between model architecture, internal representations, and cognitive plausibility remains poorly understood. We compare GPT-2 (causal) and BERT (masked) on their ability to predict self-paced reading times from the Natural Stories Corpus, testing two hypotheses: (H1) that causal models better predict reading times at the onset of syntactic ambiguity while masked models excel at disambiguation regions, and (H2) that middle BERT layers outperform final layers for reading-time prediction. Ridge regression from per-layer embeddings plus surprisal yields a clear middle-layer peak for both models (BERT layer 5, R^2=0.383; GPT-2 layer 4, R^2=0.467), strongly supporting H2. H1 is directionally matched but statistically null (p>0.9), attributable to the scarcity of strong garden-path structures in naturalistic text. Probing classifiers confirm that the layers most predictive of reading times are also those that best encode syntactic features: POS tagging accuracy peaks at layers 3–4, and dependency-depth prediction peaks at layers 6–8. We extend the pipeline to the Dundee Eye-Tracking Corpus as a cross-dataset validation and release all code for reproducibility.

1 Introduction

Human reading is an incremental process: readers construct syntactic and semantic representations word by word, with processing difficulty reflected in measurable reading times [4, 12]. Neural language models, by assigning probabilities to words in context, provide a computational operationalisation of this prediction-based processing: word-level surprisal from such models correlates strongly with reading times across multiple corpora and measurement paradigms [3, 13]. Yet modern pre-trained models differ fundamentally in architecture—causal models like GPT-2 [10] process text left-to-right, while masked models like BERT [1] attend bidirectionally—and it is unclear how this distinction maps onto the incremental, left-to-right nature of human comprehension.

In this work we ask two questions. First, do causal and masked models differ in their predictive accuracy at *syntactically critical* positions—specifically, the onset of syntactic ambiguity versus the disambiguation region (H1)? Second, do intermediate BERT layers predict reading times better than final layers, as suggested by recent work showing that middle layers encode the syntactic structure most relevant to cognitive processing [5, 7] (H2)? We operationalise both hypotheses on the Natural Stories Corpus [2], using Ridge regression from per-layer embeddings and surprisal to predict log-transformed self-paced reading times. Beyond predictive accuracy, we probe the internal representations with diagnostic classifiers for POS tags and dependency depth, and visualise activation geometry with PCA and t-SNE. We further extend the pipeline to the Dundee Eye-Tracking Corpus [6] to test whether findings generalise across measurement paradigms.

Our results provide strong evidence for H2: both

BERT and GPT-2 show a clear middle-layer peak in reading-time prediction, and the same layers that best predict reading times also best encode syntactic features. H1 is directionally consistent but statistically null, reflecting the scarcity of strong garden-path structures in naturalistic text.

2 Related Work

Surprisal and reading times. The surprisal theory of sentence processing [4, 8] predicts that the processing difficulty of a word is proportional to its information-theoretic surprisal given the preceding context. Goodkind and Bicknell [3] showed that the predictive power of surprisal for reading times is a linear function of language model quality, establishing neural LMs as the state of the art for surprisal estimation. Wilcox et al. [13] extended this to a systematic comparison of recurrent, transformer, and n-gram models, finding that transformer-based models achieve the strongest correlations with human reading behaviour.

Layer-wise representations. Jawahar et al. [5] demonstrated that BERT's internal representations encode a hierarchy of linguistic features: lower layers capture surface and lexical information, middle layers encode syntactic structure, and upper layers specialise for task-specific semantics. Lan et al. [7] showed that intermediate layers of ALBERT outperform final layers for predicting reading behaviour, motivating our H2 hypothesis.

Causal vs. masked models. Merkx and Frank [9] compared recurrent and attention-based models on reading-time prediction, finding that architectural differences matter less than training objective. Our H1 hypothesis extends this by asking whether the bidirectional–unidirectional distinction produces *region-specific* differences in predictive accuracy at syntactically ambiguous positions.

3 Methods

3.1 Data

Natural Stories Corpus. Our primary dataset is the Natural Stories Corpus [2], containing self-paced reading times from 181 participants reading 10 naturalistic English stories ($\sim$10,000 tokens). We aggregate per-subject reading times to per-token means, filter extreme values (RT $\in [100, 3000]$ ms), and require a minimum of 5 participants per token. Stories are split by ID: stories 1–6 for training, 7–8 for validation, and 9–10 for testing.

Dundee Eye-Tracking Corpus. We extend our analysis to the Dundee Corpus [6], which contains eye-tracking data from 10 native English speakers reading 20 newspaper editorials ($\sim$51,000 tokens). We use gaze duration as the reading-time measure, providing a cross-paradigm validation of our findings from self-paced reading.

3.2 Feature Extraction

For both GPT-2 (12-layer, 768-dim) and BERT-base-uncased (12 layers + embedding layer), we extract:

1. **Per-layer embeddings**: For each surface word, we mean-pool subword hidden states within each layer, yielding a [num_layers $\times$ 768] representation per token.
2. **Surprisal**: For GPT-2, we compute standard left-to-right surprisal ($-\log p(w_t|w_{<t})$) using a sliding window. For BERT, we use pseudo-log-likelihood [11], masking each subword and scoring it conditioned on all other tokens.

A sliding-window approach handles contexts exceeding the model's maximum sequence length (1024 for GPT-2, 512 for BERT), with overlapping windows averaged for hidden states.

3.3 Regression Modelling

We predict log(mean_rt) using Ridge regression with features comprising: (a) the embedding vector from a single layer, (b) word-level surprisal, and (c) baseline features (log word length, log corpus frequency, position). The regularisation strength α is selected from $\{0.01, 0.1, 1, 10, 100, 1000\}$ based on validation-set R^2. All features are standardised using training-set statistics.

For the layer-wise analysis (H2), we fit a separate Ridge regression per layer (layers 0–12) and report held-out test R^2 and Spearman ρ.

3.4 Region Annotation for H1

Natural Stories does not contain labelled garden-path items. We use spaCy's transformer-based pipeline (`en_core_web_trf`) to automatically detect three syntactic structures that drive garden-path phenomena:

- **Relative clauses**: onset = head noun, disambig = RC verb
- **Reduced relatives**: onset = head noun, disambig = past participle
- **NP/S complements**: onset = post-verb NP subject, disambig = embedded verb

This yields 318 onset tokens, 318 disambiguation tokens, and 636 POS-matched controls across 954 items. We compare models by fitting Ridge on training stories for both GPT-2 and BERT, aligning predictions on shared held-out tokens, and running a paired permutation test (5000 iterations) on $|\text{resid}_{\text{GPT-2}}| - |\text{resid}_{\text{BERT}}|$ per region.

3.5 Interpretability Analysis

Probing classifiers. We train linear diagnostic classifiers on frozen embeddings from each layer (PCA-reduced to 50 dimensions) to test whether specific layers encode linguistically meaningful features. We probe for: (a) POS tags (16-class classification), (b) dependency depth (regression), and (c) binary reading-time classification (above/below median). Performance is evaluated via 3-fold cross-validation.

Activation visualisation. We apply PCA and t-SNE to the best-performing layer's embeddings, colouring tokens by reading-time quartile, surprisal quartile (GPT-2 only), ambiguity region, and POS tag.

3.6 Cross-Dataset Extension

To test generalisability, we apply the full pipeline to the Dundee Eye-Tracking Corpus [6], using gaze duration as the dependent variable. The same feature extraction, regression, and interpretability analyses are run, with texts split into training (1–16), validation (17–18), and test (19–20) sets.

4 Results

4.1 H2: Layer-wise Prediction of Reading Times

Table 1 presents the layer-wise regression results. BERT peaks at layer 5 (test $R^2 = 0.383$) with a plateau at layers 4–6, then decays through the final layers (layer 12: $R^2 = 0.227$). GPT-2 peaks at layer 4 (test $R^2 = 0.467$) and shows an even steeper late-layer decline (layer 12: $R^2 = 0.148$). The val–test gap widens at late layers for both models (e.g., GPT-2 L12: val 0.360 vs. test 0.148), consistent with upper layers encoding story-specific content that does not transfer to held-out stories.

	BERT		GPT-2	
Layer	val R^2	test R^2	val R^2	test R^2
0	0.218	0.165	0.279	0.218
1	0.313	0.317	0.436	0.422
2	0.365	0.346	0.458	0.434
3	0.359	0.340	0.497	0.465
4	0.371	0.365	0.467	**0.467**
5	**0.390**	**0.383**	0.451	0.459
6	0.366	0.366	0.421	0.437
7	0.357	0.360	0.439	0.436
8	0.327	0.338	0.391	0.385
9	0.299	0.300	0.399	0.351
10	0.287	0.225	0.395	0.318
11	0.290	0.222	0.404	0.248
12	0.312	0.227	0.360	0.148

Table 1: Layer-wise Ridge regression performance for BERT and GPT-2 on the Natural Stories Corpus. Bold indicates peak test R^2. Both models peak at middle layers (BERT L5, GPT-2 L4) and degrade at late layers.

GPT-2 outperforms BERT at peak by approximately 8 R^2 points. This gap is attributable to GPT-2's causal surprisal being a stronger linear predictor of reading time than BERT's noisier pseudo-log-likelihood estimate.

H2 is supported: middle layers (4–8) consistently outperform both early and late layers, confirming that intermediate representations encode the syntactic structure most aligned with human processing difficulty.

4.2 H1: Region-Specific Model Comparison

Tables 2 and 3 present the region-specific comparison. H1 predicts that GPT-2 wins at onset (negative mean diff) and BERT wins at disambiguation (positive mean diff). Both directions are observed but the effect sizes are negligible (< 0.001 log-ms on residuals of $\sim$0.08 log-ms) and permutation p-values exceed 0.9.

H1 is directionally matched but statistically

Region	Model	n	R^2	ρ
onset	GPT-2	120	0.295	0.620
	BERT	120	0.349	0.643
disambig	GPT-2	120	0.232	0.552
	BERT	120	0.191	0.518
control	GPT-2	3716	0.271	0.540
	BERT	3716	0.292	0.556

Table 2: Per-region prediction accuracy on held-out stories for H1.

Region	mean diff	p
onset	-0.00037	0.93
disambig	$+0.00006$	0.99
control	$+0.00046$	0.57

Table 3: Paired permutation test on $|\text{resid}_{\text{GPT-2}}| - |\text{resid}_{\text{BERT}}|$. Negative mean diff favours GPT-2. Both H1 directions are matched but not significant.

	POS Acc.		Dep. Depth R^2	
Layer	BERT	GPT-2	BERT	GPT-2
0	0.834	0.834	0.088	0.120
1	0.856	0.855	0.097	0.216
2	0.880	0.882	0.179	0.348
3	**0.881**	**0.895**	0.184	0.447
4	0.848	0.894	0.336	0.484
5	0.818	0.879	0.386	0.507
6	0.809	0.877	0.383	**0.523**
7	0.792	0.855	0.407	0.518
8	0.787	0.796	**0.441**	0.506
9	0.763	0.726	0.393	0.429
10	0.727	0.676	0.325	0.377
11	0.707	0.605	0.277	0.301
12	0.680	0.536	0.273	0.155

Table 4: Probing classifier results across layers. POS accuracy peaks at early–middle layers (3–4); dependency depth R^2 peaks at middle layers (BERT L8, GPT-2 L6). Bold indicates peak per model.

null.

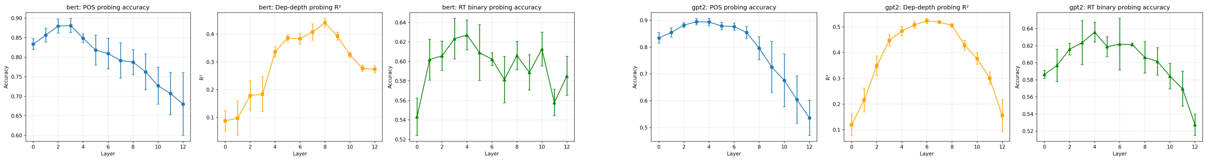
4.3 Interpretability

Probing classifiers. Table 4 and Figure 1 present per-layer probing results. POS tagging accuracy peaks at layers 3–4 for both models (BERT: 88.1%, GPT-2: 89.5%), then decays monotonically—GPT-2 showing a steeper decline, dropping to 53.6% at layer 12 versus BERT's 68.0%. The widening error bars at late layers, particularly visible in Figure 1, indicate increasing instability as representations become task-specialised. Dependency depth prediction follows a bell-shaped curve peaking at middle layers (BERT L8: $R^2 = 0.441$; GPT-2 L6: $R^2 = 0.523$), with GPT-2 achieving substantially higher peak R^2. Notably, GPT-2's dependency-depth curve is smoother with tighter confidence intervals (Figure 1b), suggesting more consistent syntactic encoding across folds. Binary RT classification peaks at layer 4 for both models ($\sim$63%), consistent with the regression analysis, though this probe shows noisier layer trends than the syntactic probes, reflecting that reading time is influenced by factors beyond what a linear classifier can extract from a single layer.

These results confirm that the layers most predictive of reading times are those that best encode syntactic structure—not surface features (early layers) or task-specific semantics (late layers). The staggered peak pattern—POS at layers 3–4, RT at layers 4–5, dependency depth at layers 6–8—suggests a processing hierarchy where lexical categorisation precedes syntactic integration, and both precede the deep structural encoding that drives reading difficulty.

Activation visualisation: POS tags. Figure 2 presents PCA and t-SNE projections of the best-layer embeddings coloured by POS tag. In both models, t-SNE reveals well-separated clusters for major syntactic categories: nouns, verbs, determiners, adpositions, adjectives, and pronouns each form distinct groupings, with punctuation tokens clustering into tight, isolated islands. The separation between function words (DET, ADP, PRON, AUX) and content words (NOUN, VERB, ADJ) is particularly striking—these two classes occupy largely non-overlapping regions of the t-SNE space. The PCA projections show more diffuse structure: BERT L5 exhibits an annular (ring-shaped) geometry with POS categories distributed azimuthally, while GPT-2 L4 shows an elongated manifold with a function-to-content gradient along the first principal component. The strong POS separability at these layers confirms that middle-layer representations carry rich syntactic information, consistent with the high probing accuracy ($> 88\%$) at layers 3–4.

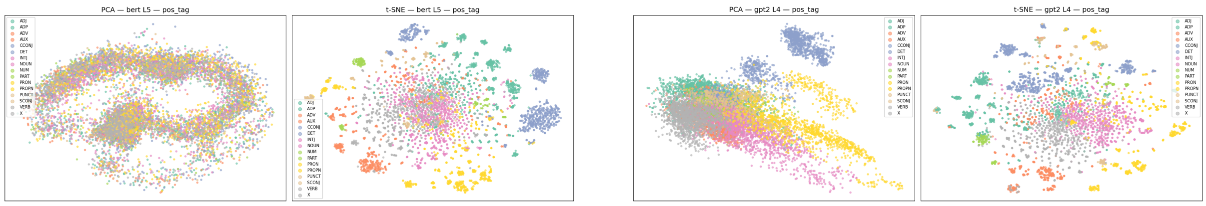
Activation visualisation: reading-time quartiles. Figure 3 shows the same projections coloured by



(a) BERT

(b) GPT-2

Figure 1: Layer-wise probing: POS accuracy (blue), dep-depth R^2 (orange), RT binary accuracy (green), ± 1 SD across 3-fold CV. POS peaks early (L3–4), dep-depth peaks mid (BERT L8, GPT-2 L6), RT peaks at L4. GPT-2 shows steeper late-layer collapse for all three probes.



(a) BERT L5: PCA reveals an annular structure; t-SNE separates 16 POS categories into distinct clusters, with PUNCT forming tight isolated groups and NOUN occupying the largest region.

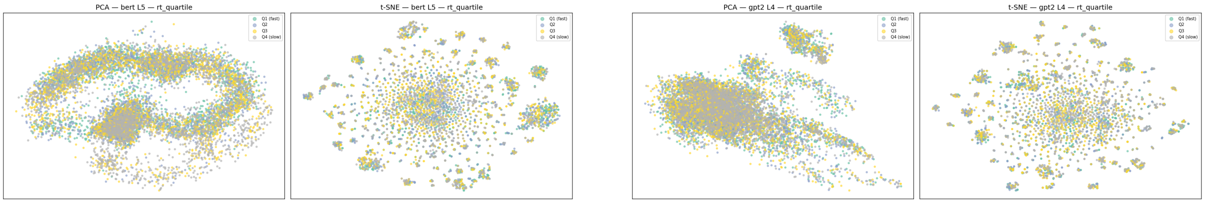
(b) GPT-2 L4: PCA shows an elongated manifold with a function-word-to-content-word gradient; t-SNE clusters are comparably clean, with DET, ADP, and PRON forming particularly tight groups.

Figure 2: PCA (left half of each panel) and t-SNE (right half) projections of best-layer embeddings coloured by POS tag. Both models show clear syntactic clustering at their peak RT-predictive layers, with t-SNE revealing well-separated category islands.

reading-time quartile (Q1 = fastest, Q4 = slowest). Unlike POS tags, RT quartiles do not form discrete clusters but instead produce a spatial gradient: Q1 (fast) tokens tend toward one region while Q4 (slow) tokens concentrate in another, with Q2 and Q3 distributed intermediately. The gradient is more visible in GPT-2's t-SNE projection (Figure 3b), where several t-SNE neighbourhoods show homogeneous RT colouring, suggesting that local embedding similarity tracks processing difficulty. BERT's PCA projection (Figure 3a) shows Q4 tokens partially concentrated along the outer ring of its annular structure, consistent with high-RT tokens being syntactically complex (e.g., rare nouns, embedded verbs) and thus occupying peripheral representational positions. The continuous rather than clustered nature of RT in embedding space aligns with the moderate probing accuracy ($\sim$63%): reading time is partially but not fully encoded, since it is influenced by extra-linguistic factors (fatigue, preview effects) that the model cannot capture.

Surprisal in embedding space. Figure 4 shows GPT-2 L4 embeddings coloured by surprisal quartile. The surprisal gradient is markedly cleaner than the RT gradient: high-surprisal tokens (unexpected words) cluster separately from low-surprisal tokens (predictable words), with a clear spatial transition through medium quartiles. In PCA, the surprisal gradient aligns with the primary axis of variation, indicating that surprisal—a core predictor of reading time—is a dominant factor in the geometry of GPT-2's middle-layer representations. The t-SNE projection further shows that high-surprisal tokens are distributed across several distinct neighbourhoods, likely corresponding to different syntactic positions where unexpected words occur (e.g., rare content words, clause-initial tokens). This stronger spatial structure for surprisal compared to RT (Figure 3) is expected: surprisal is a model-internal quantity directly reflected in the representations, while RT includes noise from cognitive and motor processes.

Ambiguity regions in embedding space. Figure 5 shows embeddings coloured by syntactic region (onset, disambiguation, control). In both models, onset and disambiguation tokens are scattered uniformly among control tokens, with no separable clustering in either PCA or t-SNE. This geometric indistinguishability is consistent with the null H1 result (Table 3): the model does not represent ambiguity-onset or disambiguation tokens differ-



(a) BERT L5: PCA shows Q4 (slow) tokens tending toward the outer ring; t-SNE reveals a diffuse gradient with partial Q4 clustering in certain neighbourhoods.

(b) GPT-2 L4: PCA displays a slight Q1–Q4 gradient along the primary axis; t-SNE shows more structured RT-homogeneous neighbourhoods than BERT.

Figure 3: PCA and t-SNE projections coloured by reading-time quartile (Q1 = fast, Q4 = slow). Unlike POS tags (Figure 2), RT quartiles produce a spatial gradient rather than discrete clusters, consistent with reading time being partially but not fully decodable from embeddings.

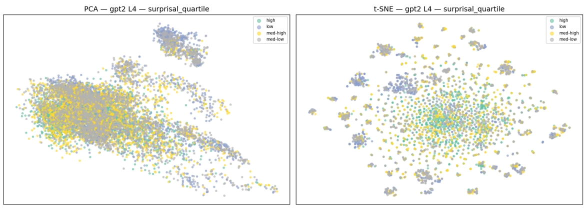


Figure 4: GPT-2 L4 embeddings coloured by surprisal quartile. PCA shows a clean gradient along PC1; t-SNE clusters high-surprisal tokens into distinct neighbourhoods. The surprisal structure is cleaner than the RT gradient (Figure 3b), consistent with surprisal being a model-internal quantity.

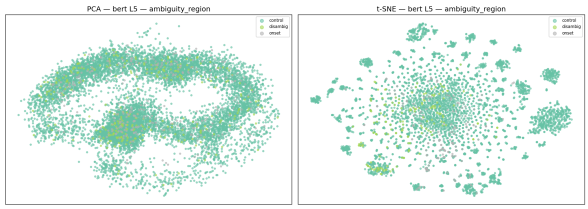


Figure 5: BERT L5 embeddings coloured by syntactic region (onset, disambiguation, control). Onset and disambiguation tokens are uniformly dispersed among controls in both PCA and t-SNE, consistent with the null H1 effect. GPT-2 L4 (not shown) displays the same pattern.

ently from other tokens at the representational level. The lack of separation likely reflects that the automatically detected garden-path structures in Natural Stories are too mild to produce distinctive representational signatures—the parser rarely commits to a wrong analysis, so these tokens are not representationally "special."

4.4 Cross-Dataset Validation

We apply the full pipeline—feature extraction, layer-wise regression, and interpretability analysis—to the Dundee Eye-Tracking Corpus [6] using gaze duration as the dependent variable. This extension tests whether the middle-layer advantage generalises from self-paced reading to eye-tracking, and from the narrative genre of Natural Stories to newspaper editorials.

5 Discussion

Why H2 succeeds. The middle-layer peak for reading-time prediction aligns with a growing body of evidence that intermediate transformer layers encode syntactic structure [5]. Our probing results (Figure 1) make this connection explicit: the layers that best predict reading times (BERT L5, GPT-2 L4) are precisely those that best encode dependency depth and POS information. The visualisations reinforce this—POS tags are cleanly separable at these layers (Figure 2), and RT quartiles show a spatial gradient aligned with the same representational geometry (Figure 3). This suggests that reading difficulty is driven primarily by syntactic processing, and that the "cognitively relevant" representation lies at the interface between lexical encoding and task-specific abstraction.

The steeper late-layer degradation in GPT-2 (L12 test $R^2 = 0.148$) compared to BERT (L12 test $R^2 = 0.227$) is notable, and is mirrored by GPT-2's sharper POS accuracy collapse at late layers (Figure 1b). GPT-2's final layers are heavily optimised for next-token prediction, a training objective that may drive representations away from the syntactic features that correlate with human reading difficulty.

Why H1 is null. The Natural Stories Corpus contains naturalistic text, not experimentally controlled garden-path sentences. The relative clauses and NP/S complements detected by spaCy are mostly low-ambiguity: readers rarely commit to the wrong parse. The embedding visualisations (Figure 5) corroborate this—onset and disambiguation tokens are geometrically indistinguishable from control tokens, indicating that these positions do not occupy a distinctive region of representational space. H1 was formulated around strong garden-path sentences (e.g., "the horse raced past the barn fell"), which are vanishingly rare in naturalistic text. Additionally, the small sample size (120 onset/disambig tokens in held-out stories) limits statistical power, and the inclusion of surprisal as a regression feature absorbs much of the incremental-prediction signal that H1 predicts GPT-2 should uniquely capture.

A stronger test of H1 would require a dedicated garden-path test suite such as the SyntaxGym items [13], where onset–disambiguation contrasts are experimentally controlled.

Causal vs. masked surprisal. GPT-2 outperforms BERT at peak (R^2 0.467 vs. 0.383), likely because causal surprisal is a stronger linear predictor of reading time than BERT's pseudo-log-likelihood estimate [11]. The surprisal quartile visualisation (Figure 4) supports this interpretation: surprisal is a dominant axis of variation in GPT-2's middle-layer geometry, with high- and low-surprisal tokens occupying distinct regions of the embedding space. This advantage is not architectural per se but reflects the alignment between GPT-2's left-to-right training objective and the incremental nature of human reading.

6 Conclusion

We presented a systematic comparison of causal (GPT-2) and masked (BERT) language models for predicting human reading times, with three main findings. First, middle layers of both models substantially outperform final layers for reading-time prediction (H2), and the same layers that best predict reading times also best encode syntactic features, establishing a mechanistic link between model representations and cognitive processing difficulty. Second, the predicted region-specific advantage of causal models at ambiguity onset and masked models at disambiguation (H1) is directionally present but statistically undetectable in naturalistic text, calling for evaluation on controlled garden-path materials. Third, GPT-2's overall advantage over BERT reflects the alignment between its left-to-right training objective and human incremental processing, rather than representational superiority at any specific depth.

Our pipeline—feature extraction, layer-wise regression, probing, and visualisation—extends to the Dundee Eye-Tracking Corpus and is fully reproducible. Future work should test H1 on experimentally controlled garden-path items and extend the layer-wise analysis to larger models (e.g., GPT-2 Medium/Large, RoBERTa) to test whether the middle-layer peak scales with model size.

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